# COMPOSITION, CANCELLATION AND DESCENT OF ETALE MORPHISMS INTO AN ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the first part through the definition and stability under composition, the second part by base change along a surjective etale chart with descent of etaleness, and the third part by base change along an arbitrary scheme over the space and a further descent step.

*Editorial note.* The label lemma-etale-over-space names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Editorial note.* Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-properties.tex –label lemma-etale-over-space –start 2236 –end 2309 –local-id 28049 returned RESULT=OK (L28049 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-spaces-properties/candidates-halfA.json.

*Editorial note.* The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

*Allowed standard prerequisites (editorial).*

- Definition of an etale morphism from a scheme into an algebraic space. Spaces, the definition of relative representable properties, together with the fact that any morphism from a scheme into an algebraic space is representable. Supplies the interpretation of etaleness into an algebraic space as etaleness after base change to any scheme.
- Permanence of etaleness under composition and base change. Morphisms, the lemmas that etale morphisms are stable under composition and under base change. Supplies the etaleness of the composites and of the base changes appearing in the three parts.
- Descent of etaleness along a surjective etale morphism. Descent, the lemma allowing the etale property of a morphism of schemes to be checked after a surjective etale base change. Supplies the conclusion in parts (2) and (3) that the morphism in question is etale.

*Source reference.* Original lemma 02KM: sources/stacks-chow/tags.txt, source lines 3377–3377. Prerequisite pointer invoked at proof line(s) 2291; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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*Date*: stacks-project:spaces-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 02GN: sources/stacks-chow/tags.txt, source lines 3234–3234. Prerequisite pointer invoked at proof line(s) 2268; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 02GW: sources/stacks-chow/tags.txt, source lines 3243–3243. Prerequisite pointer invoked at proof line(s) 2281; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original definition 025V: sources/stacks-chow/tags.txt, source lines 2828–2828. Prerequisite pointer invoked at proof line(s) 2260; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 02WK: sources/stacks-chow/tags.txt, source lines 3807–3807. Prerequisite pointer invoked at proof line(s) 2271; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 02WJ: sources/stacks-chow/tags.txt, source lines 3806–3806. Prerequisite pointer invoked at proof line(s) 2272; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $S$ be a scheme. Let $X$ be an algebraic space over $S$. Let $U$, $U'$ be schemes over $S$.*

(1) *If $U \to U'$ is an étale morphism of schemes, and if $U' \to X$ is an étale morphism from $U'$ to $X$, then the composition $U \to X$ is an étale morphism from $U$ to $X$.*
(2) *If $\varphi : U \to X$ and $\varphi' : U' \to X$ are étale morphisms towards $X$, and if $\chi : U \to U'$ is a morphism of schemes such that $\varphi = \varphi' \circ \chi$, then $\chi$ is an étale morphism of schemes.*
(3) *If $\chi : U \to U'$ is a surjective étale morphism of schemes and $\varphi' : U' \to X$ is a morphism such that $\varphi = \varphi' \circ \chi$ is étale, then $\varphi'$ is étale.*

*Proof.* Recall that our definition of an étale morphism from a scheme into an algebraic space comes from Spaces, Definition 025V via the fact that any morphism from a scheme into an algebraic space is representable.

Part (1) of the lemma follows from this, the fact that étale morphisms are preserved under composition (Morphisms, Lemma 02GN) and Spaces, Lemmas 02WK and 02WJ (which are formal).

To prove part (2) choose a scheme $W$ over $S$ and a surjective étale morphism $W \to X$. Consider the base change $\chi_W : W \times_X U \to W \times_X U'$ of $\chi$. As $W \times_X U$ and $W \times_X U'$ are étale over $W$, we conclude that $\chi_W$ is étale, by Morphisms, Lemma 02GW. On the other hand, in the commutative diagram

$$\begin{array}{ccc} W \times_X U & \longrightarrow & W \times_X U' \\ \downarrow & & \downarrow \\ U & \longrightarrow & U' \end{array}$$

the two vertical arrows are étale and surjective. Hence by Descent, Lemma 02KM we conclude that $U \to U'$ is étale.

To prove part (3) choose a scheme $W$ over $S$ and a morphism $W \to X$. As above we consider the diagram

$$\begin{array}{ccccc} W \times_X U & \longrightarrow & W \times_X U' & \longrightarrow & W \\ \downarrow & & \downarrow & & \downarrow \\ U & \longrightarrow & U' & \longrightarrow & X \end{array}$$

Now we know that $W \times_X U \to W \times_X U'$ is surjective étale (as a base change of $U \to U'$) and that $W \times_X U \to W$ is étale. Thus $W \times_X U' \to W$ is étale by Descent, Lemma 02KM. By definition this means that $\varphi'$ is étale. □